Group Project Data Guidebook: which data you may use, and how to get it approved

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Course: New Media Technology and Analytics (NMTA, 20001920076) ·**Fall 2026** ·Group project ·**Version 1.1** (Week 2) ·*Governing documents:* `project_brief.md`, `project_question_menu.md`, `project_data_dictionary.md`, *and the artificial intelligence (AI) use policy (*`ai_use_policy.pdf`*).*

Most teams use the course project dataset, the five Lumen Beauty tables, and need nothing else from this guidebook. Read Section 1, then return to your question. The rest of the guidebook is for teams whose question needs data that the course dataset does not contain.

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1. Start here: the course dataset comes first

The course project dataset is the Lumen Beauty full Sep-Nov 2026 operations export: five tables on livestream sessions, order lines, users, stream comments, and follow and share events. It is already in the `data/` folder of the project kit, and `project_data_dictionary.md` documents every column. It carries no licensing or privacy risk because it is synthetic: a generator program created it, and no real person appears in it. Every ready-made question on the menu is built on it, and it can carry most questions that teams propose themselves.

Use other data only when your question needs something the five tables do not contain, for example a second platform, a longer period, or a real brand. Other data is open to teams that choose **Your own question** on the menu, and to a ready-made question only when the instructor approves it in writing. Every team that uses other data sends a one-page data plan with its topic claim (Section 5).

Collecting your own data earns no extra marks. The rubric scores the quality of the analysis through seven lenses, and none of them rewards the effort of finding data.

2. Four tiers of data, from easiest to hardest

Move down a tier only when the tier above cannot give your question what it needs. Three terms first. A **license** is the set of terms under which the owner of a dataset lets others use it. **Personal data** is any

information that identifies a person or can be linked to one: a name, a phone number, a precise address, an account handle, a face, or a comment that can be traced back to its author. **De-identified** data has had every such field removed or replaced, for example by a customer number that leads back to no one.

Tier	Source	Use it when	Conditions
0	The course project dataset	Default: every ready-made question and most teams	None: it is in the kit and documented
1	A licensed public dataset (Section 4)	Your question needs something the five tables lack	The license allows course use; no personal data beyond de-identified codes; small enough, or sampled down, to run on a laptop
2	An official interface or a seller or creator dashboard export	Your team already has legitimate access to a platform account	The account owner's written permission; the interface's terms and request limits; no personal data leaves the account
3	Collection of non-personal data from public web pages	Tiers 1 and 2 cannot supply what the question needs	Prior written approval; the rules in Section 7

Tier 1 means downloading a dataset that its owner has published under a license that allows use in a course. Section 4 lists datasets whose licenses were checked for this course. You may propose a dataset that is not on the list if your data plan shows its license.

Tier 2 covers two routes. An **application programming interface (API)** is a documented channel through which a platform lets approved programs request its data. A **dashboard export** is a file that a seller or creator account downloads from its own back office. Either route is open only to a team that already has legitimate access, for example through an internship or a brand partnership, and only when the account owner agrees in writing that the data may be used for this course project. Follow the interface's terms and its request limits. Export only the fields your question needs, aggregated or de-identified: no buyer names, phone numbers, addresses, or account handles.

Tier 3 is collecting non-personal data from public web pages whose terms permit it. It needs the instructor's prior written approval, and Section 7 sets out what it involves. No team fetches a page with a program before its plan is approved.

3. What no tier allows

These rules hold at every tier, the course dataset included.

- **No personal data.** Do not collect names, phone numbers, precise addresses, account handles, faces, or user comments that can be traced to their author. No scraping personal data, ever.
- **No collection without permission.** Scraping without permission breaks platform terms and can break China's Personal Information Protection Law (PIPL). Collect nothing from a site whose terms do not allow it.
- **No getting around barriers.** Do not bypass a login wall, a verification puzzle that tells people from programs, a request limit, or any other barrier that a site puts in place.
- **No borrowed access.** Use no account, password, or cookie that is not yours. A cookie is a small file that a website stores in your browser to remember a login.
- **No platform user data in external AI services.** User comments, profiles, and transaction records stay out of the prompts and uploads of ChatGPT and every other external AI service, "anonymized" or not, as the AI-use policy requires. For AI help with code, describe your columns instead, or practice on the course dataset.
- **No data without a license you can show.** If you cannot find a dataset's license, you cannot use the dataset.

- **No data you cannot rerun.** Your notebook must reload the data and reproduce every number in the report on an ordinary laptop.
- **No Tier 2 or Tier 3 work before approval.** Export or collect nothing until the instructor has approved your data plan in writing.

4. Public datasets checked for this course

Every entry below was checked on 2026-09-19 against its own web page, and the license shown is the one that the source states. Licenses can change, so recheck the page before you download, and credit the dataset in the data section of your report in the way its page asks. If a page does not open on your network, use another dataset listed for the same method.

Most entries come from the University of California, Irvine (UCI) Machine Learning Repository, which needs no account. Sizes are those the pages state, in kilobytes (KB) or megabytes (MB). The entries carry five kinds of terms:

- **Creative Commons Attribution 4.0 (CC BY 4.0)** lets you use and adapt the data for any purpose if you credit its creator.
- **Creative Commons Zero (CC0)** places the data in the public domain.
- **Creative Commons Attribution-ShareAlike (CC BY-SA)** adds one condition: anything you derive from the data carries the same license.
- **Creative Commons Attribution-NonCommercial-ShareAlike 4.0 (CC BY-NC-SA 4.0)** also rules out commercial use. A course project is not commercial.
- **A publisher's own terms of service** apply to one entry, which states them in place of a Creative Commons license.

Association rules (Week 3)

- **Online Retail.** UCI; CC BY 4.0; 22.6 MB. All 541,909 transaction lines of a British online gift retailer from December 2010 to December 2011; customers appear only as five-digit numbers. It also supports customer segmentation by recency, frequency and monetary value (RFM) in Week 6. https://archive.ics.uci.edu/dataset/352/online+retail
- **Online Retail II.** UCI; CC BY 4.0; 43.5 MB. The same retailer from December 2009 to December 2011: 1,067,371 lines. https://archive.ics.uci.edu/dataset/502/online+retail+ii
- **User Behavior Data from Taobao for Recommendation (淘宝用户购物行为数据集).** Alibaba Cloud Tianchi (阿里云天池); CC BY-NC-SA 4.0; 905.80 MB download. About 100 million clicks, purchases, cart additions and favorites by about 988,000 Taobao users between 25 November and 3 December 2017. Advanced: work with a sample, not the full file. Downloading needs an Alibaba Cloud account and either student verification or the platform's approval. It also suits classification and clustering. https://tianchi.aliyun.com/dataset/649

Classification (Week 4)

- **Online Shoppers Purchasing Intention Dataset.** UCI; CC BY 4.0; 1 MB. 12,330 browsing sessions on an online shop, each labeled by whether it ended in a purchase. https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset
- **Bank Marketing.** UCI; CC BY 4.0; two files of about 1 MB together. 45,211 telephone marketing contacts of a Portuguese bank, labeled by whether the client subscribed to a term deposit. It is a marketing question but not a new-media one: use it only as a fallback. https://archive.ics.uci.edu/dataset/222/bank+marketing

Regression (Week 5)

- **Online News Popularity.** UCI; CC BY 4.0; 23.2 MB. 39,797 articles from the news site Mashable, each with 58 features and the number of times it was shared on social networks. https://archive.ics.uci.edu/dataset/332/online+news+popularity
- **Facebook Metrics.** UCI; CC BY 4.0; 37 KB. 500 posts from the Facebook page of a cosmetics brand in 2014, with their reach, impressions and engagement. https://archive.ics.uci.edu/dataset/368/facebook+metrics
- **China Statistical Yearbook 2025 (中国统计年鉴 2025).** National Bureau of Statistics of China (国家统计局);

the bureau's terms of service. National and provincial statistics, in Chinese, with no account needed. The terms let users download and use the published statistics, provided each reuse is a fair citation that names the bureau's website, www.stats.gov.cn. Use these figures only as background variables, not as the core of a new-media question. https://www.stats.gov.cn/sj/ndsj/2025/indexch.htm (terms: https://www.stats.gov.cn/wzgl/202302/t20230217_1912857.html)

Clustering (Week 6)

- **Facebook Live Sellers in Thailand.** UCI; CC BY 4.0; 357.8 KB. 7,051 Facebook posts (videos, photos, statuses and links) by 10 Thai fashion and cosmetics sellers, with comment, share and reaction counts. https://archive.ics.uci.edu/dataset/488/facebook+live+sellers+in+thailand
- **Wholesale customers.** UCI; CC BY 4.0; 14.7 KB. 440 clients of a wholesale distributor, with their annual spending in six product categories. https://archive.ics.uci.edu/dataset/292/wholesale+customers
- Online Retail, listed above, also supports RFM segmentation.

Text analysis (Week 7)

- **Sentiment Labelled Sentences.** UCI; CC BY 4.0; 82.2 KB. 3,000 review sentences from Amazon, Yelp and IMDb, each labeled positive or negative. https://archive.ics.uci.edu/dataset/331/sentiment+labelled+sentences
- **Women's E-Commerce Clothing Reviews.** Kaggle; CC0 as stated by the uploader; the page does not state the size. 23,486 customer reviews from an unnamed clothing retailer, anonymized by the uploader, with ratings and a recommend flag. Downloading needs a free Kaggle account. https://www.kaggle.com/datasets/nicapotato/womens-ecommerce-clothing-reviews

Social network analysis (Week 8)

- **fb-pages-company.** Network Repository; CC BY-SA. About 14,100 verified company Facebook pages linked by about 52,100 mutual likes, collected in November 2017. The nodes are brand pages, not people. https://networkrepository.com/fb-pages-company.php
- **com-amazon.** Network Repository; CC BY-SA. About 334,900 Amazon products, linked when customers frequently bought them together (about 925,900 links). https://networkrepository.com/com-amazon.php
- Network Repository states the license for all its data on one page, https://networkrepository.com/policy.php. Cite the repository as each dataset page asks.

Livestreaming

- The `lb_streams` table of the course dataset is the best livestream data available to you: 78 sessions with viewers, comments, shares, and gross merchandise value (GMV).
- Facebook Live Sellers in Thailand, listed under clustering, is the only public livestream-related dataset checked for this guidebook. It records posts, not whole streaming sessions.

5. The data plan you send with your topic claim

Any team that uses data other than the course dataset fills in the one-page plan below and attaches it to its topic-claim email, which is due Wed 2026-10-14 23:59 (see `project_question_menu.md`). The instructor approves or rejects each field together with your topic claim, and the decision comes with the confirmed topics in Week 5 (Thu 2026-10-15). Send the plan early: the sooner it arrives, the more time you have to fix a rejected field.

Team number: _____ ·Question: _____

#	Field	Your answer	Instructor
1	**Source.** Dataset or account name, owner, web address, and tier (1, 2 or 3)		

#	Field	Your answer	Instructor
2	**License.** Its name, and the sentence on the source page that grants use, quoted		
3	**Personal data.** Does the data contain any? If so, which fields, and how you remove them before analysis		
4	**How you obtain it.** Download; interface or dashboard export, with the account owner's written permission attached; or collection under Section 7		
5	**Size and rerun.** Rows, file size, and how your notebook reloads the data		
6	**Fallback.** Confirm in one sentence: if the data cannot be obtained, or its license or privacy check fails, you keep your question and answer it with the course dataset		

Instructor decision: approved ·approved with changes ·not approved

Comments: _____

6. Five pieces of advice

1. **Plan the fallback first.** Choose a question that the course dataset can still answer. If your outside data then fails in Week 6, you lose days, not the project.
2. **Make it rerun.** Hand in your own data with the notebook on Canvas, or hand in the script that regenerates it with a fixed random seed (the number that makes a random process repeat exactly). Web pages change, so keep a dated copy of anything you collect. For a public dataset, record the exact file you downloaded and the date.
3. **Draft the ethics section early.** Answer the checklist below before you start. The answers become the first draft of the ethics and compliance section of your report.
4. **Do not collect data to look diligent.** It earns no extra marks (Section 1). Spend the time on the analysis.
5. **Know what the teaching assistant does.** The teaching assistant helps you read a license and checks your data plan before you send it. The teaching assistant does not write collection code for you and does not supply platform accounts, passwords, or cookies.

Ethics checklist for teams using other data

- **Source.** Who created the data, and does its license allow use in a course project?
- **Consent.** Did the people in the data agree to this kind of use, or is the data aggregated or de-identified so that no one can be recognized?
- **De-identification.** Which fields could point to a person, and how did you remove them?
- **Law.** Would your collection and use satisfy PIPL and, where people in the European Union are involved, the General Data Protection Regulation (GDPR)?
- **AI use.** Did any platform user data enter an external AI service? It must not.

7. Tier 3: lawful collection, step by step

Tier 3 is the last resort, and it opens only after the instructor approves your data plan in writing. It covers one narrow case: collecting **public, non-personal** data from pages whose owner allows automated access. Prices in a public catalogue, published statistics, and official announcement pages can qualify. User

profiles, comment threads, follower lists, private messages, and anything behind a login never qualify, whatever a site's terms say.

Step 1: look for a published file first. Many organizations publish the same numbers as a file you may simply download, and a published file is not collection at all. Collect only what no file gives you.

Step 2: read the terms of use. Find the site's terms page and read what it says about automated access, robots, crawlers, or data extraction. If the terms forbid automated access, stop: no course exercise justifies going on. Quote the sentence that permits your collection in your data plan. A plan without that sentence is refused.

Step 3: read `robots.txt`. A site may publish a file named `robots.txt` at its root, for example https://www.scrapethissite.com/robots.txt, listing the paths that automated clients must not fetch. That file carries the line `Disallow: /faq/`, so a program stays away from that path even though a browser may open it. Where the file does not exist, as at `books.toscrape.com`, nothing is disallowed, and the terms of use still govern what you may do.

Step 4: collect politely. A program has to behave better than a person with a browser. Name your team in the User-Agent header, the line that tells a site which program is asking, so that the site can write to you. Fetch one page at a time and pause between requests. Stop at the first error instead of retrying in a loop. Keep the raw HyperText Markup Language (HTML) exactly as it arrived, with the date and time in the file name, and parse that saved copy: the page you fetch tomorrow is not the page you analyzed today.

The example below runs against `books.toscrape.com`, a shop built for scraping practice whose front page says "We love being scraped!". Install the two libraries once with `pip install requests beautifulsoup4`.

```
import time
from pathlib import Path

import requests
from bs4 import BeautifulSoup

# Name your team and an address the site can write to.
HEADERS = {"User-Agent": "NMTA course project (team 7, 2510096@tongji.edu.cn)"}


def fetch(url):
    """Fetch one page politely and return the bytes exactly as they arrived."""
    response = requests.get(url, headers=HEADERS, timeout=10)
    response.raise_for_status()  # stop at the first error, never retry in a loop
    time.sleep(2)                # at most one request every two seconds
    return response.content


page = fetch("https://books.toscrape.com/")

# Keep the raw page with a timestamp, then parse the saved copy.
stamp = time.strftime("%Y-%m-%dT%H-%M")
Path("raw").mkdir(exist_ok=True)
Path("raw/books_" + stamp + ".html").write_bytes(page)

soup = BeautifulSoup(page, "html.parser")
rows = [(card.h3.a["title"], card.select_one("p.price_color").get_text(strip=True))
        for card in soup.select("article.product_pod")]
print(len(rows), rows[:2])
```

Step 5: log every run. Keep a collection log: the date and time of each run, the pages fetched, how many rows came back, and which version of the script produced them. Summarize that log in the data section of your report, and hand in the raw pages with your notebook so that a reader can rerun your numbers.

What this chapter does not teach, and the course forbids. None of it applies to Douyin (抖音), Kuaishou (快手), Xiaohongshu (小红书), WeChat (微信), or Taobao (淘宝). Their terms generally forbid automated collection, and the posts and comments on them are personal data. The course therefore teaches no way around an anti-bot measure, no simulated login, no mobile app signing, and no headless browser dressed up as a person. A question that needs any of those has the wrong data source: go back to Tier 0 or Tier 1.

What a Tier 3 data plan adds. Name the pages you will fetch, quote the terms sentence that permits it, state what `robots.txt` says, give the pause between requests, and say what you will store and hand in.

Questions: email `huwei72@tongji.edu.cn`, cc `2510096@tongji.edu.cn` (the teaching assistant), with the subject line `NMTA: project data`.